

INDIA TECH CAREER INTELLIGENCE

Salary, Skills & Hiring Trends Analysis — Project Summary Report

Data Analytics Final Project

Executive Summary

This project analyzes India's technology job market using a 100,000-record dataset. The analysis covers salary patterns, job roles, technical domains, skills, experience levels, locations, work modes, applicants, hiring trends, and career growth. Python is used for data cleaning, feature engineering, EDA and statistical analysis, while Power BI is used as the interactive reporting layer.

Project Problem

India's technology job market is continuously changing across skills, roles, salaries, locations, experience requirements and work arrangements. It can be difficult to identify which technology areas have stronger representation, how salaries vary, and what hiring patterns are visible in the available job data. This project converts the raw job dataset into structured analytical findings and an interactive Power BI dashboard.

Objectives

- Analyze India's technology job market and identify important hiring trends.
- Understand salary patterns across job roles, technical domains, experience levels and locations.
- Identify frequently represented technology skills and examine their relationship with salary.
- Analyze experience, education, work mode and company characteristics in relation to career opportunities.
- Create meaningful visualizations and an interactive Power BI dashboard.
- Provide actionable insights for job seekers, recruiters and career decision-making.

Project at a Glance

100,000 Job postings	33 → 36 Columns after engineering	₹849.95K Average salary midpoint	81.51% Average career growth	6.84 Average skill count
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Technology & Workflow

Python	Pandas, NumPy, Matplotlib and Seaborn for cleaning, transformation, EDA and visualization.
Power BI	Interactive dashboard for salary, job structure, skills, locations and hiring insights.
Workflow	Problem Definition → Dataset → Initial Overview → Cleaning → Feature Engineering → EDA → Insights → Export → Power BI → Conclusion.
Domain	Information Technology / Technology / Human Resources / Career Analytics.

1. DATASET & DATA PREPARATION

From raw job records to an analysis-ready dataset

Dataset Information

Source	Mendeley Data — India Tech Career Intelligence [1M]
Dataset used	100,000 records × 33 original features
Publication	June 22, 2026 — Version 1
DOI	10.17632/h2rmbfr68t.1
License	CC BY 4.0
Dataset focus	Technology jobs, salaries, skills, locations, experience, work mode, applicants and career growth.

Initial Overview

- Initial dataset shape: 100,000 rows × 33 columns.
- Duplicate records: 0.
- Missing values were concentrated in probation_period (33,499 records).
- The dataset includes numerical, categorical, Boolean and date-related fields.
- The numerical variables include salary minimum/maximum, stipend, skill count and applicant count.

Data Cleaning & Transformation

- Filled missing probation_period values with 'Not Specified' because the absence represents an unspecified job-listing detail.
- Checked and removed duplicate records; no duplicates were present.
- Removed industry because it contained only one value: Information Technology.
- Removed location_country because it contained only India and added no analytical variation.
- Removed job_description because the project focuses on structured job, salary, skill, hiring and career attributes.
- Renamed company_size → employee_range, applicant_count → applicants, interview_difficulty → interview_level, is_active → active_job and career_growth_index → career_growth.

Outlier & Skewness Checks

- Outliers were checked for the major numerical variables using the Interquartile Range (IQR) method.
- The lower and upper bounds were calculated using $1.5 \times \text{IQR}$ to identify potential outliers.
- Salary-related variables were checked for unusually high or low observations.
- No problematic salary outliers were detected, so valid salary records were retained without removal.
- Skewness was calculated for the numerical variables to identify highly skewed distributions.
- No numerical variable showed skewness greater than 1.
- Therefore, no log transformation or other skewness correction was required.
- The numerical variables were considered suitable for further EDA and visualization.

Feature Engineering

- $\text{salary_midpoint} = (\text{salary_minimum} + \text{salary_maximum}) / 2$
- $\text{salary_range} = \text{salary_maximum} - \text{salary_minimum}$
- $\text{salary_range_percent} = \text{salary_range} / \text{salary_minimum} \times 100$
- posted_month, posted_month_name and posted_quarter were created from posted_date.
- posted_date and application_deadline were converted to datetime.

Final Prepared Dataset

After preprocessing and feature engineering, the working dataset contains 100,000 rows × 36 columns and is exported as India_Tech_Career_Intelligence_Cleaned.csv for Power BI.

2. EXPLORATORY DATA ANALYSIS

Univariate, bivariate, multivariate and statistical analysis

Univariate Analysis

Bivariate Analysis

- Career growth vs salary: correlation ≈ -0.003 , indicating almost no linear relationship.
- Average salary by work mode: On-site $\approx ₹850,779$; Hybrid $\approx ₹849,952$; Remote $\approx ₹849,525$.
- Average salary by experience: Freshers $\approx ₹850,379$; Mid-Senior $\approx ₹849,661$; difference $\approx ₹718$.
- Correlation analysis shows salary_maximum vs salary_midpoint ≈ 0.965 and salary_midpoint vs salary_range ≈ 0.861 .
- Skill count has almost no linear relationship with salary midpoint in this dataset.

Multivariate Analysis

- Education + technical domain + experience were combined with salary using heatmap analysis.
- Salary + skill count + experience were analyzed using a 10,000-record sample.
- Company size + work mode + salary were compared using grouped/heatmap analysis.
- Location + technical domain + job opportunities were examined across major states.
- Salary + skills + career growth were examined together using a 3,000-record pairplot sample.

GroupBy, Pivot & Statistical Summary

GroupBy	Used to compare counts and averages by experience, work mode, technical domain, location and other categories.
Pivot table	Used to compare salary across education, technical domain, experience and work mode combinations.
Correlation	Used to identify the strength and direction of numerical relationships.
Descriptive statistics	Used mean, median, quartiles, minimum, maximum and standard deviation to summarize distributions.

Skill Frequency Analysis

- skills_extracted contains multiple comma-separated skills per job.
- Skills were split, exploded into individual records, cleaned and counted.
- Python is the most frequently represented skill with 33,212 job postings.
- Linux appears in 33,134 job postings.
- HTML, JavaScript, React, CSS, Express, Tailwind CSS, MongoDB and Node.js each appear in 16,872 postings.
- The processed skill summary is exported as India_Tech_Career_Skills_Summary.csv for Power BI.

3. KEY FINDINGS & RECOMMENDATIONS

What the analysis communicates and how it can be used

Key Findings

- Salary:** Average salary midpoint is approximately ₹849,949, with a median of approximately ₹850,314.
- Experience:** Mid-Senior roles represent 59.9% of postings, compared with 40.1% for Freshers / Entry Level.
- Work mode:** Remote + Hybrid together account for approximately 79.7% of postings.
- Technical domains:** The six technical domains are closely balanced; no single domain dominates the dataset.
- Roles:** Cloud Infrastructure Intern, Azure Cloud Engineer and AWS Solutions Architect are among the most represented titles.
- Career growth:** Average career growth index is 81.51%, with a median of 82%.
- Salary relationships:** Salary midpoint has almost no linear relationship with career growth or skill count.
- Geography:** Maharashtra has the highest state-level posting count at 24,895.
- Applicants:** Average applicant count is approximately 247 per job; total applicants are approximately 24.69M.
- Skills:** Python and Linux are the two most frequently represented skills in the analyzed dataset.

Recommendations

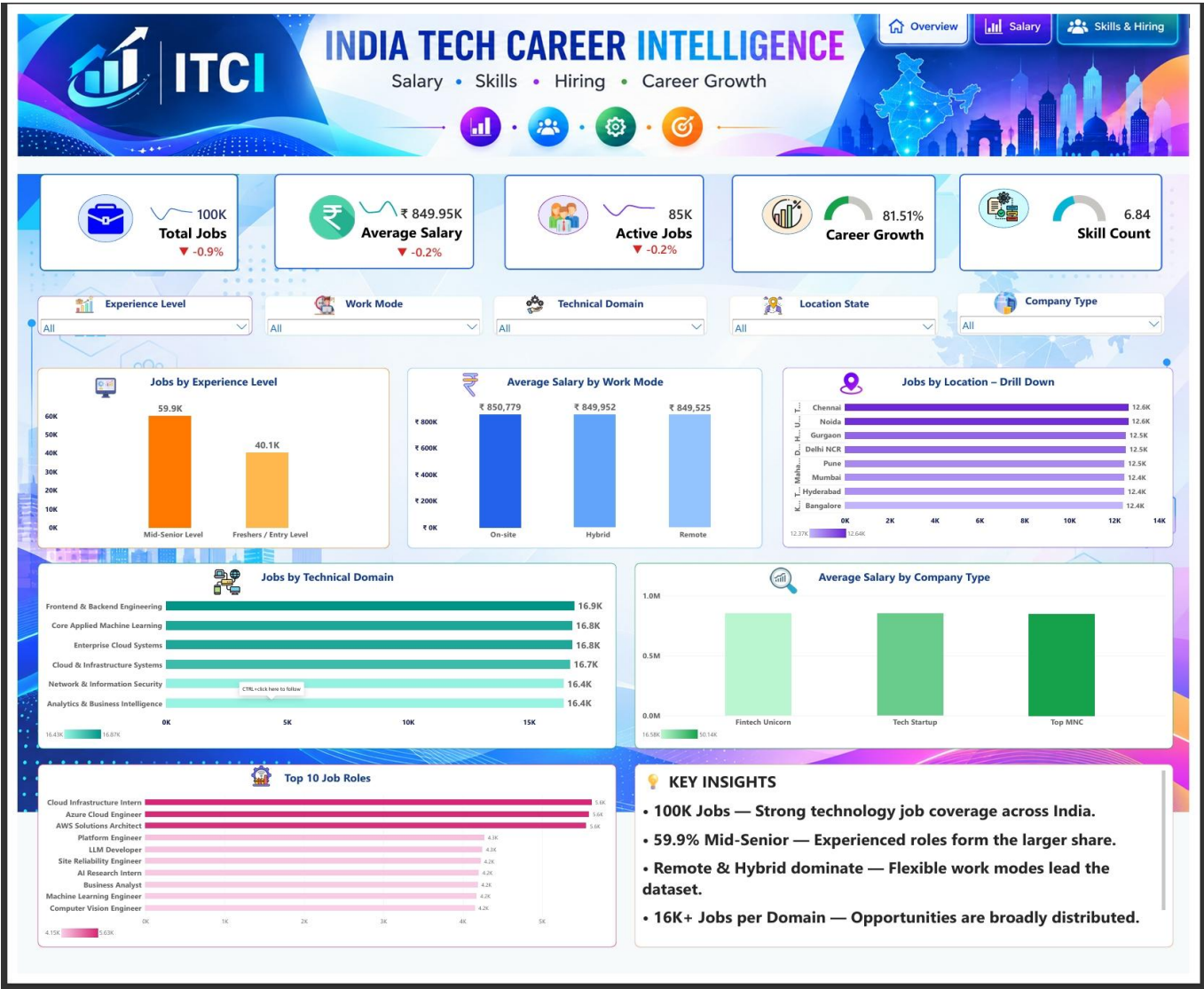
- Job seekers can prioritize frequently represented skills such as Python and Linux while building role-specific technology skills.
- Candidates should consider remote and hybrid opportunities because these work modes represent the majority of postings in the dataset.
- Recruiters can use the dashboard to compare applicant volume, salary structure and hiring representation across job categories and locations.
- Career decisions should not rely on salary alone because salary shows almost no linear relationship with the career-growth index in this dataset.
- Organizations can use location, work mode, technical domain and experience comparisons to understand the structure of their hiring market.
- The Power BI dashboard can be used interactively with slicers and drill-down to explore specific job categories, locations and career segments.

Project Deliverables

Python / Colab	Cleaned dataset, feature engineering, EDA, visualizations, groupby, pivot, correlation and statistical summaries.
Power BI	Three-page interactive dashboard: Executive Overview; Salary & Job Structure Intelligence; Skills & Hiring Intelligence.
Clean dataset	India_Tech_Career_Intelligence_Cleaned.csv
Skills summary	India_Tech_Career_Skills_Summary.csv
Documentation	This 7-page Word summary report with dedicated dashboard screenshot spaces.

4. POWER BI — PAGE 1

Executive Overview | Salary, skills, hiring and career-growth snapshot



Dashboard Summary

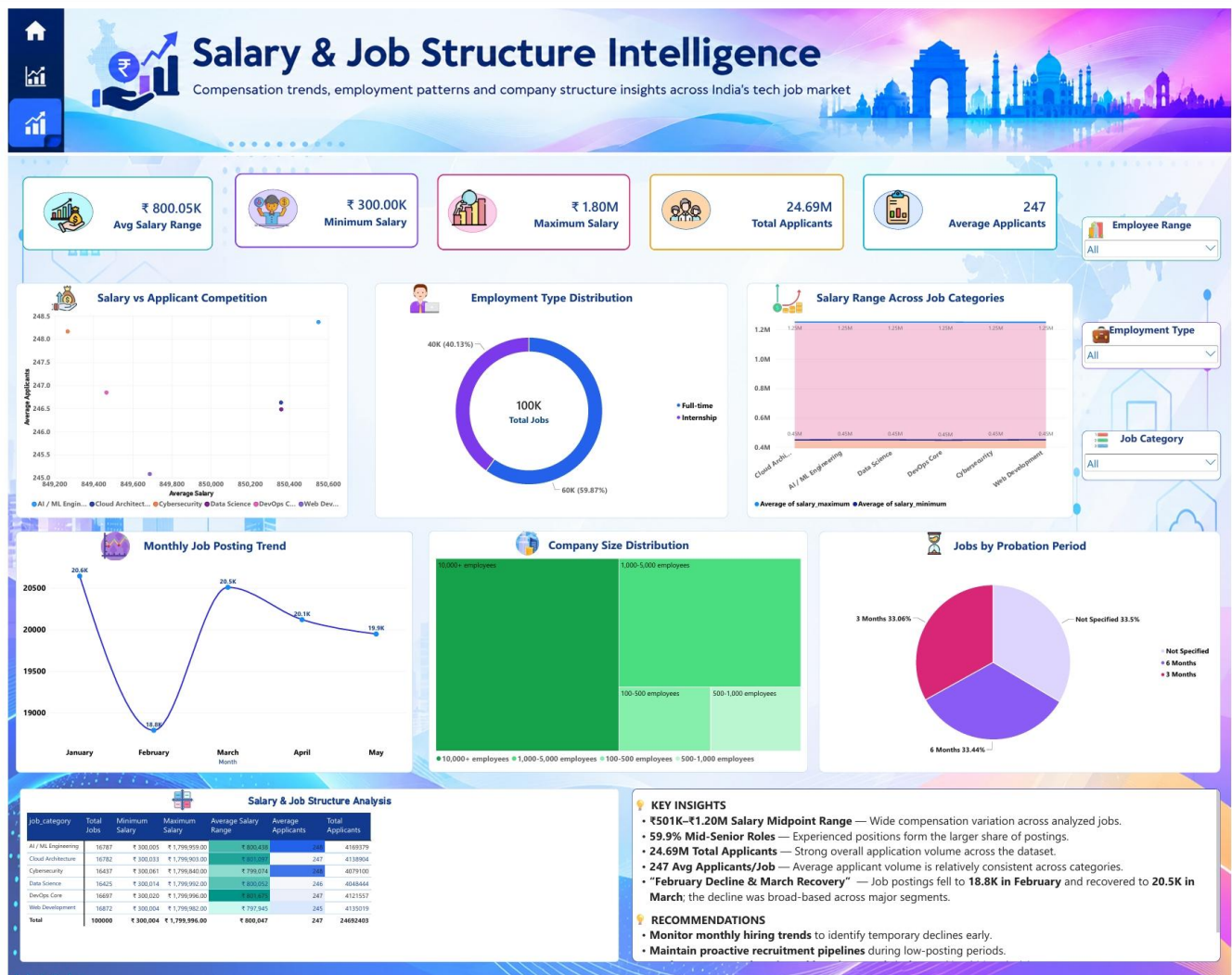
The Executive Overview provides the high-level view of the technology job market. It combines KPI cards, slicers and comparative visuals to summarize job volume, salary, active jobs, career growth and skill count. The page also supports location drill-down from state to city and job title.

Key Dashboard Insights

- 100K total job postings provide broad coverage of the analyzed technology market.
- Average salary midpoint is approximately ₹849.95K.
- 85K jobs are active in the analyzed dataset.
- Average career growth is approximately 81.51%.
- Remote and Hybrid work modes together represent approximately 79.7% of postings.
- The location visual supports drill-down analysis from Location State → Location City → Job Title.

5. POWER BI — PAGE 2

Salary & Job Structure Intelligence | Compensation and hiring structure



Dashboard Summary

Page 2 focuses on salary structure, applicant competition, employment type, company size, probation periods and monthly posting trends. It combines KPI cards with a scatterplot, donut, salary-range comparison, line chart, treemap, probation chart and an analytical matrix.

Key Dashboard Insights

- Average Salary Range is shown separately from Average Salary to represent salary spread within job postings.
- Total Applicants are approximately 24.69M, with an average of about 247 applicants per job.
- February recorded the lowest monthly job-posting volume at approximately 18.8K, followed by a recovery to about 20.5K in March.
- The February decline was broad-based across job categories, technical domains, major locations and experience levels; therefore, the dashboard does not support attributing the decline to one specific segment..
- The salary-range visual compares minimum and maximum salary levels across job categories.
- The matrix provides detailed job count, salary and applicant comparisons by job category.

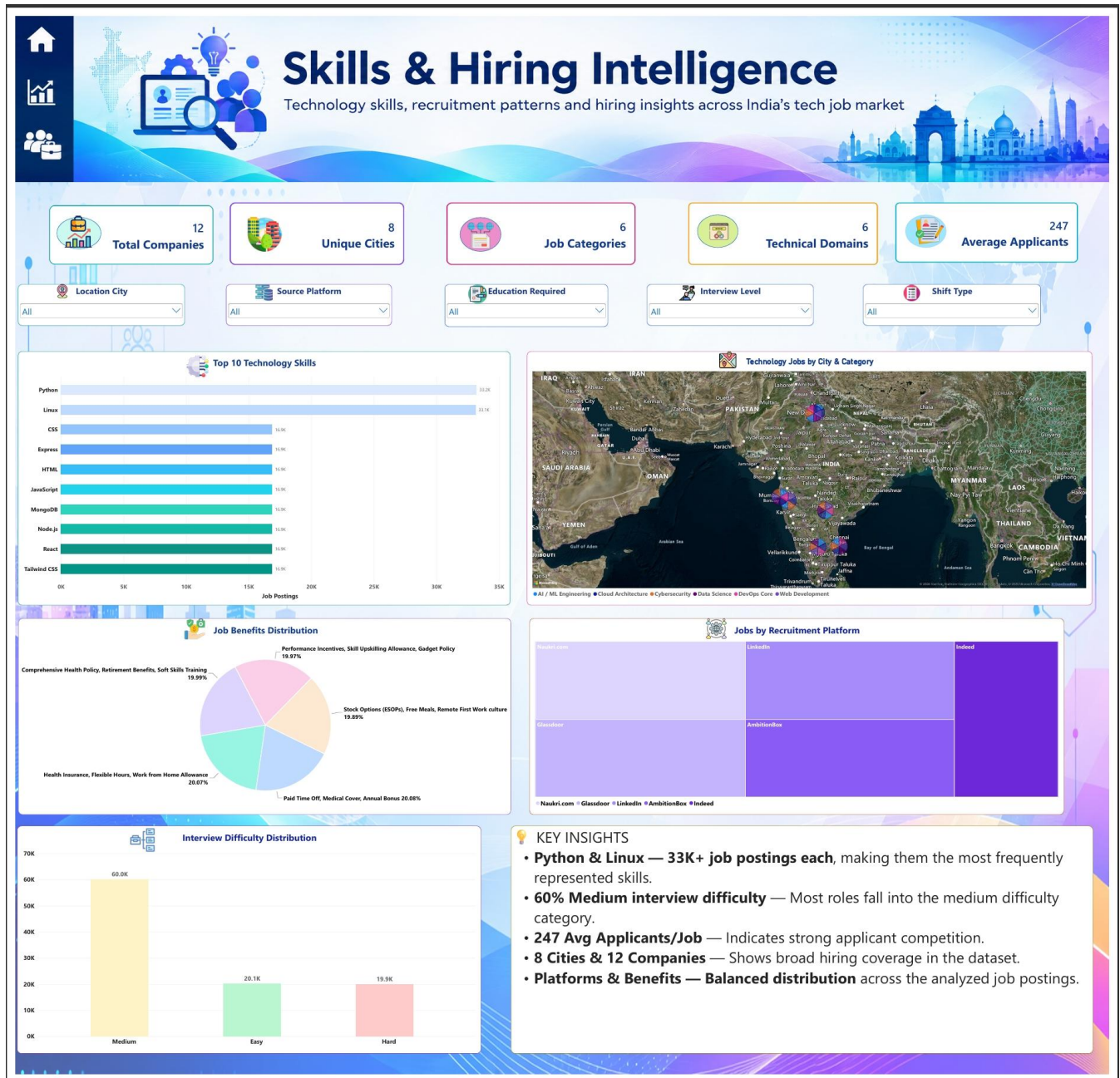
Recommendations

- Monitor monthly job-posting trends to identify temporary declines and recoveries early.
- Maintain proactive recruitment pipelines during low-posting periods so that hiring activity does not depend only on monthly fluctuations.
- Compare job category, technical domain, location and experience-level trends before making hiring decisions.

- Use salary and applicant metrics together when evaluating recruitment competitiveness rather than relying on salary alone.
- For a definitive explanation of the February decline, combine this dataset with additional external information such as seasonal hiring patterns, recruitment activity or broader market indicators.

6. POWER BI — PAGE 3 & CONCLUSION

Skills & Hiring Intelligence | Skills, locations, platforms and recruitment patterns



Dashboard Summary

Page 3 focuses on technology skills and recruitment patterns. It combines Top 10 Technology Skills, city-level mapping, benefits, recruitment platforms and interview difficulty, supported by slicers for location, source platform, education, interview level and shift type.

Key Dashboard Insights

- Python and Linux are the most frequently represented skills, with more than 33k postings each.
- The map provides city-level and category-level hiring visibility.
- Interview difficulty is concentrated in the Medium category.
- Recruitment platforms and job benefits show broadly balanced representation in the analyzed dataset.

- The page provides a practical view of the skills and recruitment side of the technology job market.

Conclusion

The project successfully follows an end-to-end data analytics workflow from problem definition and dataset selection through cleaning, feature engineering, exploratory analysis, statistical summaries, insight generation and interactive Power BI reporting. The results provide a structured view of salary, skills, hiring patterns, experience, locations, work modes and career-growth indicators in the analyzed technology job dataset.

Project source: Mendeley Data — India Tech Career Intelligence [1M] | DOI: 10.17632/h2rmbfr68t.1

GitHub: github.com/bharathisri1991-code/India-Tech-Career-Intelligence-Salary-Skills-Hiring-Trends-Analysis-Using-Python-And-Power-BI